

ZIXUAN KE

Website ◇ Github ($\approx$ 1k stars) ◇ Google Scholar ($\approx$ 3k citations, 25 h-index) ◇ zke4.uic@gmail.com

BIOGRAPHY

I am a senior research scientist at Salesforce AI Research with 30+ papers (20+ first-authored) at top venues (NeurIPS, ICLR, etc.), with multiple oral presentations, a best paper nomination, and an outstanding survey award. My research builds **autonomous agentic systems** for an **ever-changing world** characterized by emerging domains, events, tools, skills, and agents. This spans three interconnected themes: (1) **Reasoning and Agentic Systems**: designing RL-based frameworks for agent orchestration, inference-time scaling, and multi-step reasoning; (2) **LLM Post-training**: adapting LLMs to new domains and tasks via RL, SFT, and continual pre-training while preserving general capabilities (1k+ citations); and (3) **Continual Learning**: enabling knowledge transfer and preventing catastrophic forgetting across tasks and domains (1k+ citations). My earlier work also includes **NLP** (sentiment analysis, dialogue) and **argument mining** (essay scoring, persuasiveness).

INDUSTRY EXPERIENCE

Salesforce AI Research, Senior Research Scientist

May. 2026–Present, Palo Alto, CA

Salesforce AI Research, Research Scientist

May 2024–May 2026, Palo Alto, CA

Agents, Reasoning and Post-training (RL, SFT, CPT). → 12 publications; 5 first-authored; 2 orals, 1 best paper nomination, 1 outstanding survey award

- Led **MAS-Orchestra** [ICML’26]: built RL training framework and function-calling abstraction for holistic multi-agent orchestration, composing entire MAS at each step rather than incrementally. Also built **MAS-Bench**, a controlled benchmark studying when MAS outperforms single-agent systems. Outperforms GPT-5 and Claude-Sonnet-4.5 by up to 23% across 5 benchmarks with 10× efficiency gain.
- Led **MAS-Zero** [SEA@NeurIPS’25, **Oral**]: built inference-time self-refinement framework for automatic MAS design with zero supervision, achieving Pareto-frontier performance across reasoning, coding, and agentic tasks (up to 16.7% accuracy gain).
- Led **LLM Reasoning Survey** [TMLR’25, **Outstanding Survey Award**; NeurIPS’25 Tutorial]: survey on inference scaling, learning to reason, and agentic systems.
- Led **FinDAP** [EMNLP’25, **Oral**, Best Paper Nomination]: built end-to-end post-training pipeline (CPT+SFT+DPO) for 7B–32B models, including synthetic data generation (FinTrain) and evaluation suite (FinEval), achieving SOTA across financial tasks. 15k+ HuggingFace downloads. Also delivered **NAACL’25 Tutorial** as sole speaker (3-hour tutorial on LLM post-training).
- Mentored 7 publications: **SkillOrchestra** [#2 HuggingFace Daily]: skill-based agent routing outperforming SOTA RL orchestrators by 22.5% with 700× cost reduction; **MAS-ProVe** [ICML’26]: analysis of process verification for multi-agent systems; **SPARKLE** [NeurIPS’25]: analytical framework for RL in math reasoning; **LiveResearchBench** [ICLR’26]: benchmark for deep research requiring long-horizon reasoning and tool use, evaluating 17 frontier systems; **FaithEval** [ICLR’25]: 4.9K-problem contextual faithfulness benchmark; **OrchRM**: reward modeling for orchestration; **IluMAS**: A cost-controlled study plus benchmarks that stress-test the multi-agent advantage.

Google DeepMind, Research Intern

Summer 2023, Mountain View, CA

Retrieval-augmented Generation. Mentors: Weize Kong, Cheng Li, Mingyang Zhang and Qiaozhu Mei
Proposed preference-aligned retrieval for RAG, bridging the gap between retrievers and LLMs. → ACL’24

Meta AI, Research Intern

Summer 2022, Menlo Park, CA

Continual Conversational Summarization. Mentors: Haoran Li, Wenhan Xiong and Asli Celikyilmaz
Developed sub-network discovery with soft-masking for continual learning of mixed tasks. → EMNLP’23

Amazon Science, Applied Scientist Intern

Summer 2021, Seattle, WA

Multi-domain Imbalanced Learning. Mentors: Mohammad Kachuee and Sungjin Lee

Proposed domain-aware contrastive knowledge transfer for multi-domain imbalanced data. → Preprint

EDUCATION

Ph.D., University of Illinois at Chicago

Aug. 2019 - May 2024

Ph.D. in Computer Science. GPA 4.0/4.0. Advisor: Bing Liu

LLM Post-training and Continual Learning

M.Sc., The University of Texas at Dallas

Aug. 2017 - Jun. 2019

M.Sc. in Computer Science. Advisor: Vincent Ng

Argument Mining and Automatic Essay Scoring

B.Sc., South China Agricultural University

Aug. 2013 - Jun. 2017

B.Sc. in Computer Science

SELECTED PUBLICATIONS

Full list on Google Scholar

Reasoning and Agentic Systems (Research Hub)

- [1]  **MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning Through Holistic Orchestration and Controlled Benchmarks**
Zixuan Ke, Yifei Ming, Austin Xu, . . . , Semih Yavuz, Caiming Xiong, Shafiq Joty
International Conference on Machine Learning (ICML), 2026
- [2]  **MAS-Zero: Designing Multi-Agent Systems with Zero Supervision**
Zixuan Ke, Austin Xu, Yifei Ming, Xuan-Phi Nguyen, Caiming Xiong, Shafiq Joty
SEA@NeurIPS, 2025. **Oral**
- [3]  **A Survey of Frontiers in LLM Reasoning: Inference Scaling, Learning to Reason, and Agentic Systems**
Zixuan Ke, Fangkai Jiao, Yifei Ming, . . . , Silvio Savarese, Caiming Xiong, Shafiq Joty
NeurIPS Tutorial, 2025; *TMLR*, 2025. **Outstanding Survey Award**
- [4]  **MAS-ProVe: Understanding the Process Verification of Multi-Agent Systems**
Vishal Venkataramani, Haizhou Shi, Zixuan Ke, . . . , Hao Wang, Shafiq Joty
International Conference on Machine Learning (ICML), 2026
- [5]  **SkillOrchestra: Learning to Route Agents via Skill Transfer**
Jiayu Wang, Yifei Ming, Zixuan Ke, Shafiq Joty
Preprint, 2026. **#2 Paper of the day on huggingface**
- [6]  **Dissecting Mathematical Reasoning for LLMs Under Reinforcement Learning**
Jiayu Wang, Yifei Ming, Zixuan Ke, Caiming Xiong, Shafiq Joty
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [7]  **LiveResearchBench: A Live Benchmark for User-Centric Deep Research in the Wild**
Jiayu Wang, Yifei Ming, . . . , Zixuan Ke, Caiming Xiong, Shafiq Joty
International Conference on Learning Representations (ICLR), 2026
- [8]  **FaithEval: Can Your Language Model Stay Faithful to Context, Even If “The Moon is Made of Marshmallows”**
Yifei Ming, Shiva Purushwalkam, Sachin Pandit, Zixuan Ke, Caiming Xiong, Shafiq Joty
International Conference on Learning Representations (ICLR), 2025

LLM Post-training (Research Hub)

- [9]  **Demystifying Domain-adaptive Post-training for Financial LLMs**
Zixuan Ke, Yifei Ming, Xuan-Phi Nguyen, Caiming Xiong, Shafiq Joty
Empirical Methods in Natural Language Processing (EMNLP), 2025. **Oral, Best Paper Nomination**

- [10]  **Adaptation of Large Language Models**
Zixuan Ke, Yifei Ming, Shafiq Joty
NAACL 2025 Tutorial. **Sole Speaker; 3-hour tutorial on LLM post-training**
- [11]  **Bridging the Preference Gap between Retrievers and LLMs**
Zixuan Ke, Weize Kong, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Michael Bendersky
Association for Computational Linguistics (ACL), 2024
- [12]  **Continual Pre-training of Language Models**
Zixuan Ke*, Yijia Shao*, Haowei Lin* and Bing Liu
International Conference on Learning Representations (ICLR), 2023
- [13]  **Adapting a Language Model While Preserving its General Knowledge**
Zixuan Ke, Yijia Shao, Haowei Lin, Hu Xu, Lei Shu and Bing Liu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022a
- [14]  **Continual Training of Language Models for Few-Shot Learning**
Zixuan Ke, Haowei Lin, Yijia Shao, Hu Xu, Lei Shu and Bing Liu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022b
Continual Learning
- [15]  **Continual and Contrastive Learning of Aspect Sentiment Classification Tasks**
Zixuan Ke, Bing Liu, Hu Xu and Lei Shu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2021
- [16]  **Continual Learning of a Mixed Sequence of Similar and Dissimilar Tasks**
Zixuan Ke, Bing Liu and Xingchang Huang
Advances in Neural Information Processing Systems (NeurIPS), 2020
- [17]  **Achieving Forgetting Prevention and Knowledge Transfer in Continual Learning**
Zixuan Ke, Bing Liu, Nainzu Ma, Hu Xu and Lei Shu
Advances in Neural Information Processing Systems (NeurIPS), 2021

PROFESSIONAL ACTIVITIES

Invited Talks & Classes:

- LLM Reasoning and Agents
 Guest Lecture at Rice University, April 15, 2026
 Talk at CAMEL, Jan 23, 2026
- Adaptation of Large Language Models
 Tutorial at NAACL (Sole Speaker), May 3, 2025
 Talk at Visa Research, Mar 4, 2025
- Adapting Large Language Models for the Dynamic World
 Talk at Snowflake, Feb 1, 2024
 Talk at Salesforce AI Research, Jan 11, 2024
 Talk at Google DeepMind, Nov 9, 2023
- Continual Pre-training in Language Models , Talk at ContinualAI, Remote, April 27, 2023.
- Continual Learning in NLP. Tutorial at DEIM23, Remote, March 6, 2023
- Lifelong and Continual Learning. A Short PhD Course, Aalborg University, 2022.

Services:

- Area Chair: NeurIPS, ICML, ACL Rolling Review
- PC Member: ICLR, NeurIPS, ICML, ACL Rolling Review, TPAMI, TKDE, among others